

The Reward as a Hypergraph: Conflict, Simplification, and a Causal Test on the Galambos Coin-Toss

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Abstract

A two-arm robot that had *never* grasped a coin — through a week of curriculum, exploration tuning, a fingertip-distance fix, and a grasp-gate — begins grasping the moment its reward is *simplified* from ~ 11 – 14 terms to four (grasp-fraction $0.10 \rightarrow 0.615$). We explain this by treating the reward as a **signed hypergraph** over its terms, where an edge sign is how two terms co-move along a grasping trajectory. The complex reward is a *frustrated* signed graph (Harary frustration index 2); the simplified one is balanced (0). A causal test — re-adding two strongly-conflicting terms — drops grasping back to 0.333, confirming that **conflict on the reward hypergraph causes the grasp failure**. The operative quantity is the *conflict magnitude* (negative-edge weight), of which the frustration index is a stricter, topological special case. The result is stated entirely in the holonomy language of signed graphs, making it native to the HyMeKo control-substrate program.

1 The breakthrough: simpler reward, real grasping

The Galambos task requires two planar arms to corral an out-of-band coin and deliver it to a zone. The long-standing failure was *knocking, not grasping*: deliveries were the coin shoved in with the fingers never both touching it. Reward *shaping* (a strict grasp-gate, settle/still terms, a fingertip-distance correction) did not fix it. The fix was the opposite move — **simplification**. Reducing `galambos_task.hymeko`’s reward bundle from ~ 11 – 14 terms to four — **approach**, **both·5** (strong contact), **zone**, **oob** — raised the grasp-fraction of deliveries from 0.10 to **0.615** (Fig. ??; MLP, 25k steps, difficulty 0.3, 50 episodes). Eight genuine two-finger grasps replaced one.

2 The reward as a signed hypergraph

A HyMeKo `reward_spec` is already a hypergraph: a hub with weighted arcs to term nodes, i.e. $R = \mathbf{w}^\top f(\text{terms})$, a hub-convolution over one hyperedge. We make its *internal* structure explicit. Along a grasping policy’s trajectory we log each term’s value $f_i(t)$ and sign each term pair by the correlation of their increments,

$$s_{ij} = \text{sign}(\text{corr}(\Delta f_i, \Delta f_j)),$$

+ if the terms rise and fall together (jointly satisfiable), – if one’s gain is the other’s loss. This yields a **signed graph** over the terms. Its **Harary frustration index** — the minimum number

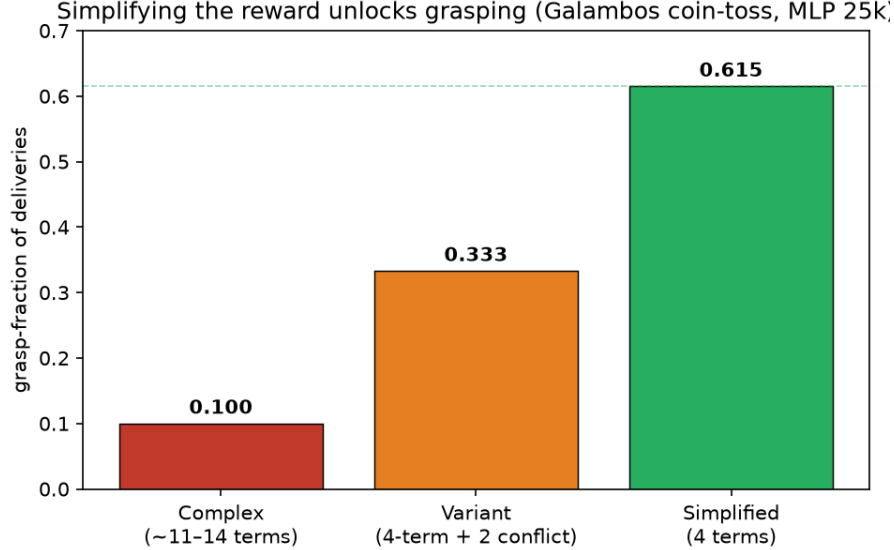


Figure 1: Grasp-fraction of deliveries against reward complexity. The complex reward knocks; the four-term reward grasps. The variant (Sec. ??) re-introduces conflict and grasping falls back.

of edges to flip for a conflict-free 2-colouring — is the irreducible conflict; index 0 means balanced, i.e. every cycle of terms has trivial holonomy and the terms split into two jointly-satisfiable groups.

A grasping probe is essential: under *random* exploration the contact/zone terms never fire and the graph is trivially balanced (underpowered). We therefore probe with a trained grasping policy, where the load-bearing terms are active.

3 Measured: the complex reward is frustrated

reward	active terms	negative edges	frustration index	balanced?
complex (11-term)	9	8	2	no
simplified (4-term)	4	1	0	yes

Table 1: Frustration of the term graph along a grasping trajectory. The dominant conflicts in the complex reward: $\text{arm_motion} \times \text{joint_velocity} = -0.24$ (anti-stall vs. smoothness, opposed by definition) and $\text{grasp_approach} \times \text{arm_motion} = -0.20$ (approaching the coin vs. generic keep-moving).

The complex reward is frustrated (index 2): a dozen small shaping/penalty terms pull against the contact-seeking the task needs. The simplified reward is balanced. The single sharpest conflict is almost tautological — *anti-stall* (keep the arm moving) versus *smoothness* (move less) cannot both be satisfied.

4 Causal test: reconstruct the conflict, lose the grasp

We re-added exactly the two conflicting terms `arm_motion` and `joint_velocity` to the balanced four-term reward and retrained. The behavioural prediction holds decisively: **grasp-fraction** $0.615 \rightarrow 0.333$ (four grasps lost, knocks up). Crucially, the *structural* prediction failed in an instructive way: the frustration index stayed 0, because two conflicting edges form a *path* (`grasp_approach`–`arm_motion`–`joint_velocity`), not an odd cycle, and a path is always 2-colourable — even though the pairwise conflict is now *stronger* (`grasp_approach` $\times$ `arm_motion` $= -0.45$).

The refined law. Grasping tracks **conflict magnitude** — the total negative-edge weight on the reward hypergraph — not the frustration index, which is the stricter special case present only when conflicts close into odd cycles (Fig. ??). Three measured points are monotone: more conflict, less grasp. The Berge-cycle framing pointed at exactly the right object (the signed term graph); the predictive invariant on it is the continuous conflict weight, not the discrete cycle count.

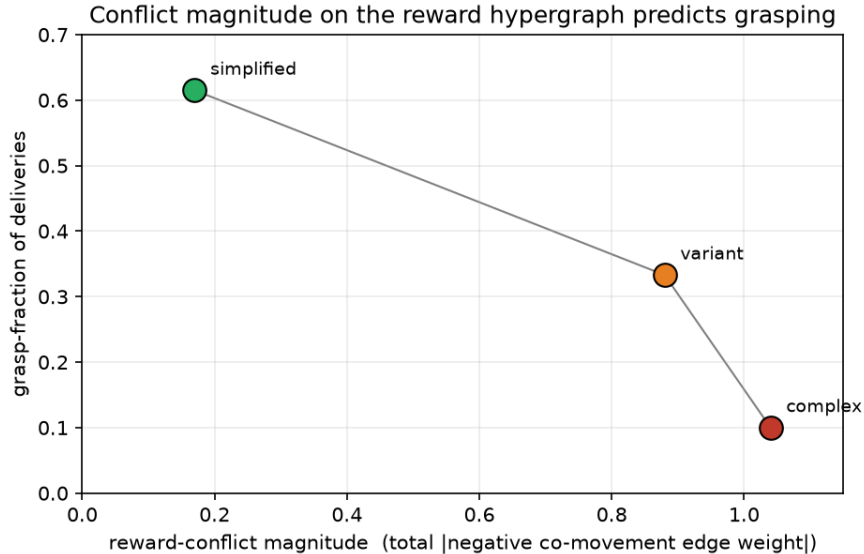


Figure 2: Grasp-fraction against reward-conflict magnitude (total |negative co-movement edge weight|). Monotone: conflict $\uparrow \Rightarrow$ grasp $\downarrow$.

5 Synthesis and next steps

Three things are now established and measured: (1) simplification unlocked grasping ($0.10 \rightarrow 0.615$); (2) reward conflict is a *computable* property of a `.hymeko` reward — a signed-hypergraph invariant, not a hunch; (3) reconstructing the conflict causally degrades grasping ($0.615 \rightarrow 0.333$). Because the reward is a hypergraph, the conflict structure is now a first-class, cheap-to-probe object.

Principal-axes study (in progress). The term co-movement matrix admits an eigendecomposition: its principal axes are the dominant modes of reward interaction (which terms are redundant, independent, or opposed), and the smallest eigenvalue of the signed Laplacian *is* the algebraic frustration. This turns reward design into a spectral problem on the hypergraph — and matters most

for the higher-ceiling but design-fragile HSiKAN controller, where a balanced, well-conditioned reward can be *guaranteed before training* rather than found by trial.

All numbers measured: MLP actor, SAC, 25k steps, difficulty 0.3, 50 evaluation episodes, seed 1. Reward conflict measured along a 12k-trained grasping policy’s trajectory. Artifacts under **reports/overnight/**. No **CORE.YAML** items touched.